__/__/____

Assignment → 4

Section A

Question 1

SRAM	DRAM
→ Static Ram : stores data using flip flops, data remains stable as long as power is on.	Dynamic RAM : stores data using capacitors, data fades and requires refreshing
→ Have very fast speed	→ Slower speed due to refresh and read-write cycles
→ No refresh needed	→ Requires periodic refresh
→ SRAM are expensive	→ DRAM are cheaper
→ SRAM have low density	→ DRAM have high density.
→ Have higher Power Consumption	→ Have lower power per bit
→ Is more Stable	→ Is less stable

Question 2 → Locality of references is the principle that a program tends to access a small portion of memory repeatedly over a short period of time, it explains why cache memory works effiuently, it has 2 types :→ temporal locality and spatical locality.

Temporal locality → time based, Recently accessed data is likely to be accessed again soon.

Spatical Locality → Space based If a memory location is accessed, nearby location are likely to be accessed soon.

Question ③ → Cache memory is used to reduce average memory access time by storing frequently used data closer to the CPU, It improves system performance by minimizing access to slower main memory

Question ④ The main cache indexing mechanisms are Direct Mapped fully associative and set associative. Direct mapping assigns one fixed location, associative allows placement anywhere and set associative is a combination of both.

Que → ⑤ Cache associativity refers to the no. of cache location where a memory block can be placed. Higher associativity reduces conflict misses but increases complexity.

Que → ⑥ → A large cache reduces miss rate and improves performance, but it also increases cost and access latency.

Que → ⑦ → Write-through → updates both cache and main memory simultaneously.
Write-back → updates only cache, main memory is updated when the block is replaced.

Question 3 → AMAT (Average memory Access time) is the average time to access memory.

formula → AMAT = Hit time + (miss rate × miss Penalty)

Hit time → time to access cache
Miss Rate → fraction of accesses that miss cache
Miss Penalty → time to fetch data from main memory

Que (9) → LRU (least Recently used) replaces the block that has not been used for the longest time, improving cache efficiency.

Que → (10) → Memory interleaving is a technique where multiple memory modules are accessed simultaneously to increase bandwidth and speed.

Que → (11) → Tracks → circular paths on the disk surface.
Sectors → sub division of tracks that store data blocks

Que → (12) flash memory is non-volatile memory used in SSD, USB drives and memory cards. It is faster than magnetic disks and has no moving parts

Que (14) → memory - mapped I/O uses the same address space for memory and I/O devices, allowing devices to be accessed using normal memory instructions

Que → (15) → I/O - mapped I/O uses a seperate address space for I/O devices, accessed using special I/O instruction

Que (16) → A programmed I/O, the CPU continuously checks the device and actively controls data transfer leading to CPU waiting.

Que (17) → In interrupt - driven I/O, the device interrupts the CPU when it is ready, improving CPU utilization by avoiding constant waiting.

Que → (18) → DMA allows devices to transfer data directly between memory and I/O without CPU involvement improving speed and efficiency.

Section B :)

Que →

Registers
↓
Cache
↓
Main Memory (RAM)
↓
Secondary storage (HDD/SSD)

⇒ Registers → Fastest memory inside CPU, stores immediate data for processing

⇒ Cache memory → very fast, small-sized memory, stores frequently used data

⇒ Main memory → larger but slower than cache, stores currently running programs

⇒ Secondary storage → Permanent storage, slowest but largest capacity

Que → 2

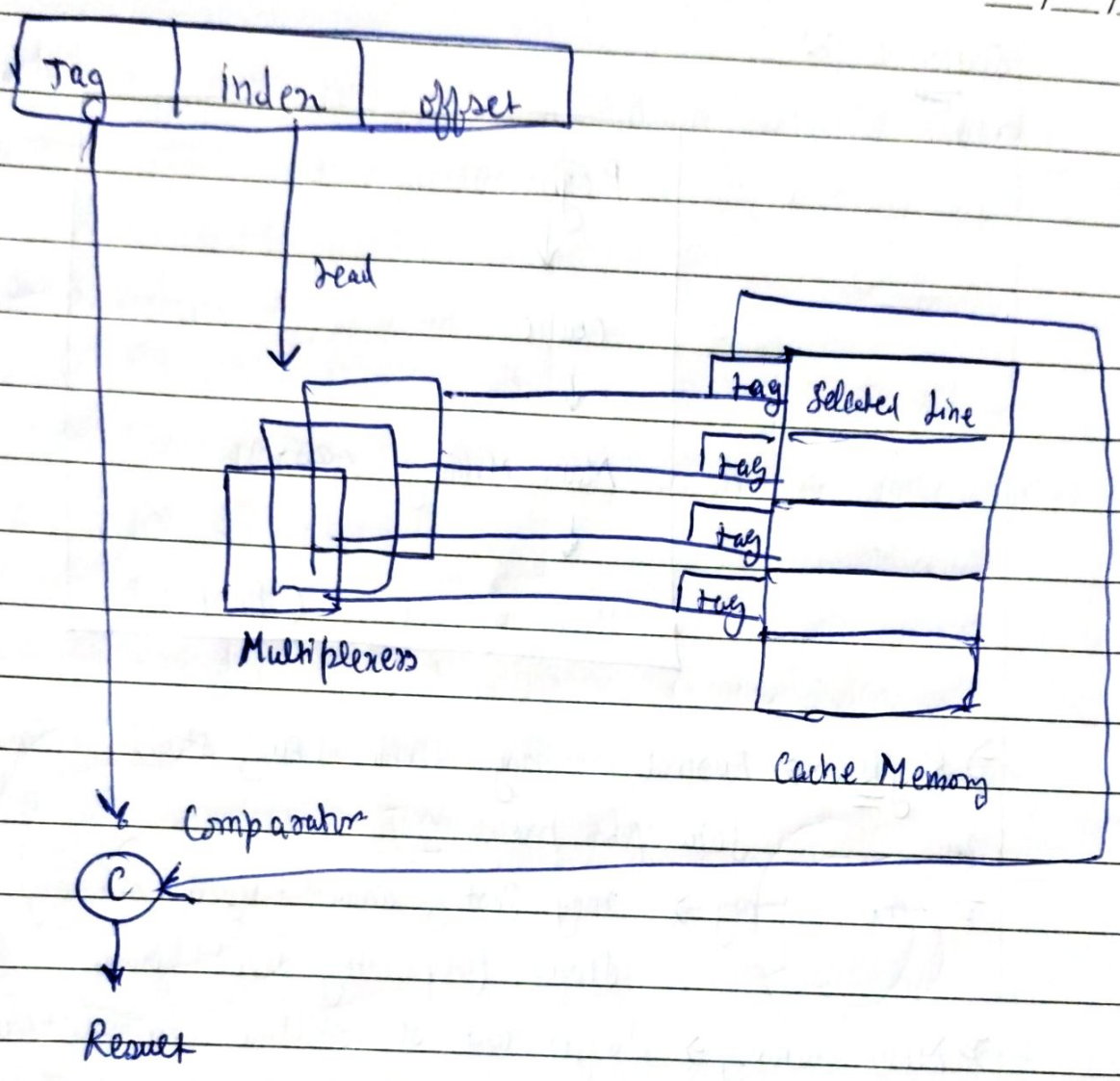
Main memory address

Tag	Index	offset

offset → select byte/word inside block

Index → Selects Cache line

Tag → Compared with stored tag

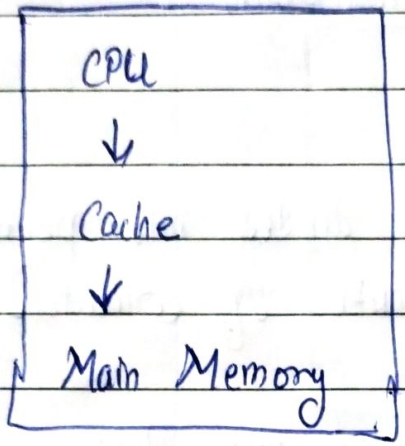


tag → identifies whether the required block is present in the selected cache line
compared with stored tag determine hit or miss

Index → selects the specific cache line,
in direct mapping each block maps to one fixed line

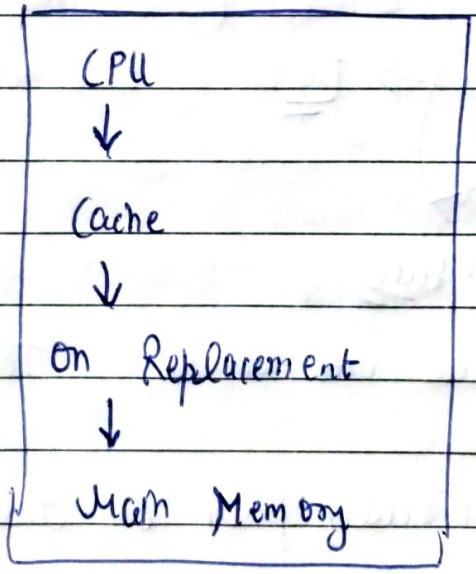
offset → selects the exact data inside the block

Que → Write - through cache



Working → CPU writes data → cache
At same time → main memory is also updated

Write back cache



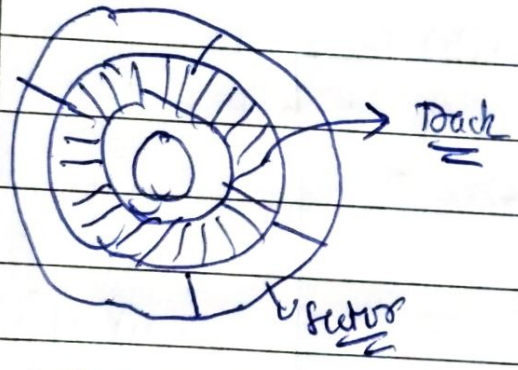
CPU writes data than only cache, main memory is updated later

In write through cache data is written to both cache and main memory simultaneously in write back cache data is written only to cache and updated in main memory when the block is replaced improving performance

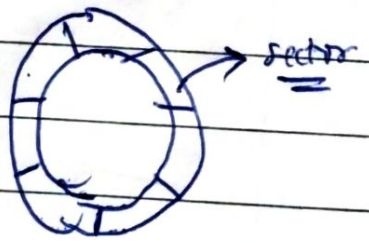
Que → Magnetic disk is a storage device that is used to write, rewrite and access data.

Architecture :→

① The entire disk is divided into platters
② Each platter consists of concentric circle called as tracks
③ these tracks are further divided into sectors which are the smallest division in the disk



Disk divided into tracks



Track divided into sectors

tracks are circular path on the disk surface
Data is stored along these rings

Sectors Tracks are divided into smaller units called sectors each sector stores a fixed amount of data.

__/__/____

Section (C) →

Que → Temporal locality applies because the same memory location is accessed multiple times within a short period of time.

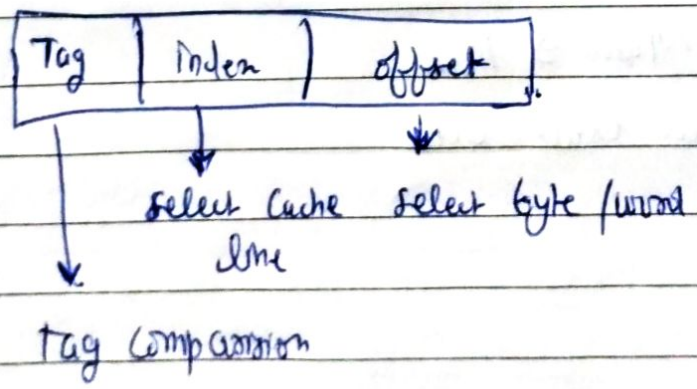
Que → Larger cache blocks increase miss penalty and cause cache pollution where unnecessary data replaces useful data, reducing overall performance

Que → In a write-through cache, data is updated in both cache and main memory simultaneously during every write operations

Que → Direct Memory Access is used, where the device transfer data directly to memory without CPU intervention.

Section (D) :→

Que



Cache memory

Index →

Line	Tag	Data
Line	Tag	Data
Line	Tag	Data
--		
Line n-1	Tag	Data

→ Address is divided into :

Index → Select cache line

tag → Checks if block is present

offset → Selects data within block

→ CPU generates address

→ Index selects line

→ Tag → compared

↳ Match → Hit

↳ No match → Miss

→ Offset → Select exact data

Que → AMAT → Hit time + (Miss Rate × Miss Penalty)

Hit time → time to access cache
Miss Rate → fraction of accesses that miss
Miss Penalty → time to fetch block from Main memory

factors affecting memory performance →

- Cache size → larger (lower miss rate but higher latency)
- Block size → larger (better spatial locality but higher miss penalty)
- Associativity → higher (fewer conflict misses but slower access)
- Replacement policy → LRU better than random